Cover letter

Dear Team/Professor,

I am quite curious about the PhD position in Starlab as it aligns with my vision to improve personalized treatment. I am inspired by Neuroelectric contributions (I even thought of contacting for opportunities). For my research work I got inspired from a recent paper that intrigued me with a question that clinicians are selecting montaged for network level TDCS but why not there are network input and montage average current output. I'm curious, so I reached out to Dr. Gaute Einevoll, University of Oslo. In our short meeting, he gave me an insight of the practical difficulties and technical insight. Its mathematically possible, but he did say, it should be done as collaborative work by researchers like us along with the clinical team.

From being an aspiring mathematician, why I chose computational neuroscience? My Masters thesis on "FFT Based Smart watches for Disabled Person" intrigued me by the idea of mathematics for healthcare. I have presented my master thesis in a conference on Feb 2024 were my idea is to use FFT (Fast Fourier Transformation) algorithm in NPU (Neural Processing Unit) integrated smartwatch so it can be of great use to disabled person as a naive Math grad all I know is how to apply mathematics but I don't know as researcher. My paper got rejected for publication. By 2025, specialized NPU chips like the Shenju W30 (marketed as the industry's first dedicated NPU smartwatch chip) began appearing. Now, ultra-efficient chips like the Ambiq Atomiq were released, designed specifically to run "Transformer" models (like ChatGPT/Gemini style AI) directly on your wrist with minimal battery drain. I was extremely happy that my idea had potential but lacked technical, practical and research insight.

Although I'm not into studying for grades, I'm always fascinated by the depth and elegance of Mathematics. During my IIT Bombay internship I loved mathematical stimulations using Geogebra software where I found my passion for research. The main root cause of my thesis idea is to personalize the needs and treatment of differently abled people. This drive got me into a Junior Research Fellow (JRF) position at AI4BNT lab at NIMHANS during my last semester in Apr 2024. I chose Neuroscience mainly because the brain is an organ that is not fully understood (mysterious). In other words, brain disease, mental health and all brain related problems need personalized solutions.

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Dr. Rajan Kashyap is my Principal Investigator, great and kind mentor. I worked with him and have contributed to the understanding of the data in the paper titled "Longitudinal evaluation of common and unique brain-networks in variants of primary progressive aphasia" published on Alzheimer's Research & Therapy (2025) and co-authored manuscript on early prediction of dementia with MCSA dataset using Brain Age Modeling methods like Gaussian Process Regression titled "Brain Ageing and Cardiometabolic Risk Factors: Prioritising Blood Pressure Control for Dementia PreventionBrain Ageing and Cardiometabolic Risk Factors: Prioritising Blood Pressure Control for Dementia Prevention" submitted in The Journal of the Alzheimer's Association is under review. I got exposed to both technical and clinical work that is the main reason I applied to the Starlab because while working on both sides I learned both clinical (like handling patients, interpreting patient data, taking assessments, giving TDCS treatment, etc..) and experienced programming skills (Coding in Python and Matlab) .

I felt knowing customer needs (that is patient and clinician problems) and updating the tool accordingly aids better research and contribution value. Now, I have contributed for the upgraded version of Individual - Systematic Approach For tDCS Analysis (iSATA) toolbox to network level as MATLAB and Python based tool released on March 2026. In addition to my technical background, I enjoy experimental and clinical work. In close collaboration with Dr. Sagarika Bhattacharjee, CNPN Lab. I lead a data collection study focused on personalized dose modeling in Transcranial Direct Current Stimulation (TDCS) for PPA-FTD patients, using the iSATA software developed by Dr. Rajan Kashyap . The training and experimental phase of this clinical study gave me a huge opportunity to know patient and clinician requirements which inspired the new version of iSATA software. I want this opportunity to expand my experience and knowledge in personalized treatment I strongly feel this consortium is the right place for me to grow as a researcher and a future entrepreneur.

Regards,

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